

SoleCart – Smart Shoe E-Commerce & Inventory Management System

1. Project Definition

SoleCart is a web-based smart shoe e-commerce and inventory management platform for selling shoes online. Customers can browse products, search using filters, add items to a cart, place orders, manage wishlists, review products, and track deliveries. Administrators and inventory staff can manage products, stock, orders, reports, and analytics. AI recommendations and n8n automation improve customer experience and reduce manual work.

2. Problem Definition

Traditional shoe stores struggle with manual inventory tracking, delayed stock updates, poor sales monitoring, and inefficient order management. Customers often experience limited search options, no personalized recommendations, and poor order tracking. SoleCart addresses these issues through automation, centralized inventory, AI recommendations, and an online shopping platform.

3. User Problems

Customers:

- Difficulty finding suitable shoes.
- Limited search and filtering.
- No personalized recommendations.
- Slow checkout.
- Poor order tracking.

Store Management:

- Manual stock tracking.
- Overstock/low-stock issues.
- Difficult order management.
- Delayed customer communication.
- Weak sales analytics.

4. Functional Requirements

- User Registration/Login
- Forgot Password & Email Verification
- Product Listing
- Product Search & Filters
- Category & Brand Management
- Product Variants (Size/Color)
- Product Images
- Wishlist
- Cart
- Checkout
- Address Management
- Order Management
- Inventory Management
- Review & Rating
- Return Request
- Coupon Management
- Invoice Generation

- Payment (Demo)
- Notifications
- Reports
- AI Recommendations
- Admin Dashboard

5. Non-Functional Requirements

Fast response, secure authentication, encryption, scalability, reliability, availability, mobile responsiveness, browser compatibility, logging, backup & recovery, maintainability.

6. Main Modules

Authentication, Customer, Admin, Product, Product Variant, Product Image, Category, Cart, Wishlist, Order, Shipping, Inventory, Payment, Notification, Analytics, AI Recommendation, Automation.

7. Users / Actors

Admin: Controls the complete system.

Customer: Registers, shops, tracks orders, manages profile, reviews products.

Inventory Staff: Updates stock, monitors inventory, generates stock reports.

8. User Role-Based Access

Admin:

- Manage users
- Manage products, variants, images
- Manage categories & brands
- Manage inventory
- Manage coupons
- Manage orders & returns
- View reports & analytics
- Send notifications

Customer:

- Register/Login
- Browse & search products
- Filter by brand, size, color, price, gender
- Wishlist
- Add to cart
- Checkout
- Manage addresses
- Track orders
- Submit reviews
- Request returns

Inventory Staff:

- Update stock
- Manage product availability
- Monitor low stock
- Generate inventory reports
- View inventory history

9. System Workflow

1. Register/Login
2. Browse products
3. Search & filter
4. Receive AI recommendations
5. Add to wishlist/cart
6. Checkout
7. Place order
8. Inventory updates automatically
9. Confirmation email & invoice
10. Delivery
11. Review & rating

10. Product Categories

Running Shoes, Sneakers, Casual Shoes, Formal Shoes, Sports Shoes, Sandals, Slippers, Boots.

11. Product Filters

Brand, Size, Color, Price, Category, Gender.

12. Automation Using n8n

Order confirmation emails, invoice emails, inventory update, low-stock alerts, abandoned cart reminders, welcome emails, promotional emails, weekly sales reports, review reminders.

13. AI Features

Personalized recommendations, similar shoes, frequently bought together, trending products, customer behavior analysis, recently viewed products.

14. Project Boundaries

Included: online shopping, inventory management, order management, AI recommendations, email automation, reviews, coupons, returns.
Not Included: real payment gateway, GPS tracking, mobile app, advanced ML models.

15. Technology Stack

Frontend: HTML, CSS, JavaScript, React.
Backend: PHP REST API.
Database: MySQL.
Automation: n8n.

16. Database Tables

users, roles, categories, brands, products, product_variants, product_images, inventory, cart, cart_items, wishlist, addresses, orders, order_items, payments, reviews, notifications, coupons, returns, reports, activity_logs.

17. Project Objectives

Provide smart online shopping, improve customer experience, automate business processes, simplify inventory management, reduce manual work, improve sales.

18. Conclusion

SoleCart combines modern e-commerce, inventory management, AI recommendations, and n8n automation into one scalable solution that improves customer satisfaction and business efficiency.